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Predicting and Identifying Low Ground Water Level Areas: A Machine Learning Approach

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***Abstract*— Groundwater depletion is an escalating concern, especially in regions where water scarcity threatens agriculture, drinking water supply, and long-term sustainability. Traditional monitoring techniques often fall short in providing timely and location-specific insights for groundwater management. This study proposes a machine learning-based framework to predict and identify low groundwater level areas in Krishna (erstwhile) district of Andhra Pradesh, using historical bi-seasonal (May and November) groundwater level data from 2019 to 2023. Three regression models—Support Vector Regression (SVR), Random Forest (RF), and Gradient Boosting (GB)—were trained on normalized data and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2. Among these, the Gradient Boosting model achieved the best performance, with an RMSE of 2.34, MAE of 1.20, and R^2 of 0.99, indicating high predictive accuracy. Predicted 2024 values were compared against historical averages to detect potential depletion zones. The results were spatially visualized in QGIS using village-level geolocation data. This approach offers a scalable and data-driven tool for proactive groundwater assessment, enabling more informed decision-making for water conservation in vulnerable regions.**

Keywords*— Groundwater Prediction, Machine Learning, Gradient Boosting, Regression models, Water resource management

I. INTRODUCTION

Groundwater plays a critical role in sustaining agriculture, domestic needs, and industrial activities, particularly in regions facing water scarcity. In India, the growing pressures of over-extraction, inadequate recharge, and unsustainable practices have led to alarming declines in groundwater levels, threatening both water security and agricultural productivity. Monitoring and predicting groundwater dynamics is therefore essential, yet it remains a significant challenge due to spatial and temporal variability, as well as the influence of multiple interrelated factors. Traditional approaches, relying on physical models and extensive field measurements, are often resource-intensive and lack scalability.

Recent progress in machine learning offers a promising alternative by enabling the analysis of large historical datasets to capture complex, nonlinear patterns in groundwater systems. Such data-driven approaches can improve forecasting accuracy, provide timely insights, and support informed decision-making. In this study, advanced regression-based machine learning techniques are applied to bi-seasonal groundwater data collected over several years. The developed models are systematically evaluated and employed to predict future groundwater levels, which are subsequently mapped to identify regions vulnerable to depletion. By combining predictive modeling with geospatial visualization, this research seeks to contribute a scalable and efficient framework for sustainable groundwater resource management and proactive conservation planning.

A. Motivation

Sustainable management of groundwater resources has become increasingly vital in the face of rising water scarcity, agricultural dependence, and the impacts of climate variability. Conventional monitoring methods, while valuable, are often constrained by high costs, limited spatial and temporal resolution, and difficulties in modeling the complex, nonlinear behavior of groundwater systems. In contrast, machine learning approaches provide a powerful and cost-effective alternative by leveraging historical data to capture intricate patterns and improve prediction accuracy. These methods enable timely and reliable insights that can guide proactive measures for groundwater conservation, reducing the risks posed by over-extraction, land use changes, and shifting climate conditions. By employing models such as Random Forest, Support Vector Regression, and Gradient Boosting, this study seeks to advance groundwater forecasting capabilities, thereby strengthening water security and supporting broader sustainable development objectives.

B. Problem Statement

Groundwater is a crucial resource for domestic, agricultural, and industrial needs, yet rising demand and unregulated extraction have led to its alarming decline. Traditional prediction methods struggle to capture the complex interactions of climatic, geological, and human factors that influence groundwater dynamics, resulting in

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limited forecasting accuracy. The absence of reliable and timely predictions hampers effective management and conservation efforts. By applying advanced machine learning techniques to historical groundwater data, this study seeks to develop accurate and practical predictive models that can guide decision-making and support sustainable groundwater management in vulnerable regions.

C. Objectives

- Develop and validate machine learning models for accurate prediction of groundwater levels using historical data.
- Identify regions at risk of groundwater depletion and represent findings through spatial visualizations to support informed water management.
- Deliver a scalable, data-driven framework that contributes to sustainable groundwater resource planning and conservation.

II. Literature Review

Groundwater has emerged as one of the most vital natural resources in supporting domestic, agricultural, and industrial activities, particularly in monsoon-dependent regions like India. The growing imbalance between groundwater extraction and recharge, coupled with rapid urbanization, has intensified the need for advanced groundwater monitoring and prediction systems. Traditional methods, although widely used, are often resource-intensive and unable to deliver accurate, large-scale temporal insights.

Recent advancements in data-driven modeling have proven effective in addressing these limitations. Evans et al. [1] introduced a web-based groundwater level mapping tool that facilitates sustainability assessments, highlighting its utility in groundwater resource management. Husna et al. [2] demonstrated the use of artificial neural networks for groundwater level prediction, emphasizing the effectiveness of machine learning (ML) in hydrological modeling. Sun et al. [3] developed data-driven models for accurate groundwater level prediction, showcasing their practical significance in water management. Ghobadi and Kang [4] presented a systematic review of ML applications in water resources, highlighting the growing adoption of AI techniques in hydrology.

Ahmed et al. [5] reviewed machine learning approaches applied to water resource management and discussed their future potential. Pham et al. [6] applied various ML algorithms in a drought-prone area to predict groundwater levels, achieving promising results. Tao et al. [7] provided a comprehensive review of groundwater prediction using ML, underscoring the strengths of algorithms like LSTM in capturing temporal dependencies. Kenda et al. [8] used statistical modeling for surface and groundwater prediction, reinforcing the importance of hybrid methods. Roy et al. [9] employed a heterogeneous ensemble of ML algorithms to improve forecasting of groundwater fluctuations. Finally, Thakur et al. [10] applied long short-term memory (LSTM) networks for groundwater level prediction, effectively modeling non-linear temporal trends.

Recent advancements in data-driven modelling have proven effective in addressing these limitations. Chen et al. [11] applied deep learning methods to predict groundwater levels, demonstrating improved accuracy over traditional statistical models. Sapitang et al. [12] evaluated various machine learning algorithms for monthly water level prediction, highlighting the potential of ML models in incorporating groundwater as an input parameter. Li et al. [13] used machine learning techniques to support sustainable irrigation in water-scarce regions, emphasizing the importance of predictive models in optimizing groundwater usage. Malakar et al. [14] leveraged GRACE satellite data for regional-scale groundwater level prediction, showing that integrating remote sensing with ML enhances spatial prediction capabilities. Zarafshan et al. [15] compared multiple ML models for groundwater prediction in the Najafabad region, illustrating the superiority of certain algorithms in capturing local hydrogeological variability.

Collectively, these studies underscore the growing adoption of machine learning and data-driven approaches in groundwater modelling, offering scalable, accurate, and cost-effective solutions for resource management and sustainable water planning.

A. Research Gap

While machine learning has advanced groundwater prediction, key gaps remain. Many studies rely on complex, multivariate datasets, making them less applicable in data-scarce regions. Seasonal variations, especially between pre- and post-monsoon periods, are often overlooked, leaving critical groundwater dynamics underexplored. Moreover, although advanced models such as LSTMs achieve high accuracy, their complexity and limited interpretability hinder practical use. There is also a lack of research on simpler models that use only long-term groundwater level data, which could provide more accessible and scalable solutions for sustainable management.

III. Proposed System

A. System Architecture

This study follows the workflow shown in Fig. 1 for predicting groundwater levels in 2024. The process involves collecting historical groundwater data, training the predictive model, forecasting future values, saving the results in a CSV file, and visualizing them using QGIS for spatial analysis.

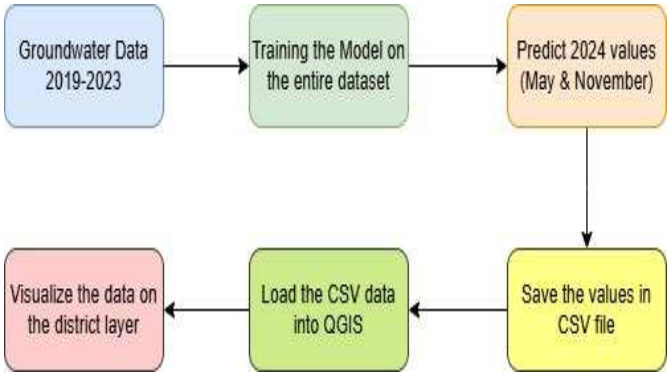


Fig. 1. System Architecture

B. Data Collection

The dataset for this study was sourced from the Ground Water and Water Audit Department, Government of Andhra Pradesh. It comprises groundwater level measurements (in meters below ground level, mbgl) recorded across 108

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villages in the erstwhile Krishna District between 2019 and 2023. Each entry includes the location details (district, mandal, and village) along with groundwater levels measured during May and November of each year.

C. Data Processing

To simplify the dataset, location details (district, mandal, and village) were merged into a single identifier. Groundwater levels from 2019 to 2022 were used as input features, while the levels for May and November 2023 served as target variables. Standardization was applied to ensure consistent scaling of features for effective model training. Records with missing values were excluded to maintain data integrity and reliability of the analysis.

D. Data Split for Training and Evaluation

The groundwater dataset, containing measurements from 108 villages between 2019 and 2023, was partitioned into training and testing subsets. A standard 80:20 split was applied, with 80% of the data used to train the models and the remaining 20% reserved for testing and performance evaluation.

E. Model Training and Evaluation

To accurately forecast groundwater levels, three widely used machine learning models were employed: Random Forest (RF), Support Vector Regression (SVR), and Gradient Boosting (GB). These models were trained using historical groundwater data collected over the past five years from 108 villages. The training process involved splitting the data into training and validation sets to ensure the model could generalize well to unseen data.

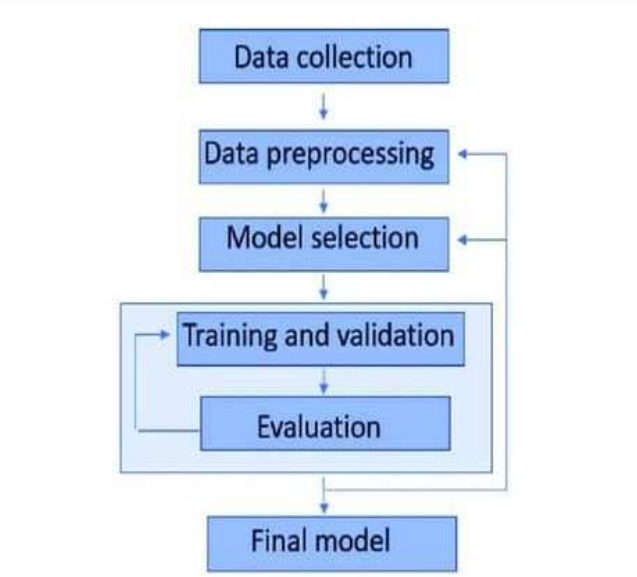


Fig. 2. Implementation Workflow

The workflow for predicting groundwater levels in 2024 is shown in Fig. 2. Using groundwater records from 2019 to 2023, we trained three models and evaluated them using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). Gradient Boosting showed the best performance and was selected for final predictions.

F. Model Retraining and Prediction

To enhance predictive accuracy, the selected Gradient Boosting model was retrained on the complete five-year dataset, leveraging all historical patterns and seasonal fluctuations. The retrained model was then used to forecast groundwater levels for May and November 2024, representing pre-monsoon and post-monsoon conditions.

Predicted levels for each of the 108 villages were compiled into a structured CSV file for post-processing and analysis. Villages with predicted groundwater levels significantly lower than historical averages were identified as potential risk zones. These areas can be prioritized for interventions such as water conservation measures, artificial recharge, or regulation of groundwater extraction.

G. GIS Integration and Visualization

To make predictions actionable, the forecasted groundwater data were imported into QGIS, a widely used open-source geographic information system. Graduated color symbology was applied to generate spatial maps, visually distinguishing villages with deeper water tables from areas with relatively stable groundwater levels.

These GIS-based visualizations allow policymakers, local authorities, and water resource managers to quickly identify critical depletion zones and plan targeted interventions. The integration of machine learning predictions with GIS ensures that the insights are not only numerical but also spatially actionable, supporting sustainable groundwater management.

IV. Results And Discussion

Among the three machine learning models tested, the Gradient Boosting model delivered the most reliable predictions, achieving the lowest error values (MAE and RMSE) and the highest R^2 score.

Table 1 shows the evaluation results of the Gradient Boosting model. The low RMSE and MAE indicate reduced prediction error, while the high R^2 value confirms strong agreement between predicted and observed groundwater levels. It was therefore selected for groundwater level forecasting.

Table 2 presents a comparative evaluation of SVR, RF, and GB models based on RMSE, MAE, and R^2 metrics. The results indicate that the Gradient Boosting model achieves superior performance, demonstrating the lowest prediction errors and the highest coefficient of determination.

Using the retrained model, groundwater levels were predicted for May and November 2024 across 108 villages in the Krishna (erstwhile) district. The results showed seasonal variations, with deeper groundwater levels during May (pre-monsoon) compared to November (post-monsoon), reflecting the natural recharge and withdrawal cycle of the region.

Comparison of predicted values with historical averages identified potential stress zones, where groundwater levels are lower than usual. These areas can be prioritized for preventive interventions to avoid water scarcity.

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The predicted groundwater levels were mapped at the village level using GIS, highlighting regions with deeper water tables. These visualizations provide policymakers with clear, actionable insights, supporting targeted measures such as artificial recharge, conservation practices, and awareness programs.

Overall, the results demonstrate the effectiveness of machine learning in forecasting localized groundwater conditions and underscore the value of integrating predictive models with geospatial analysis for sustainable water resource management.

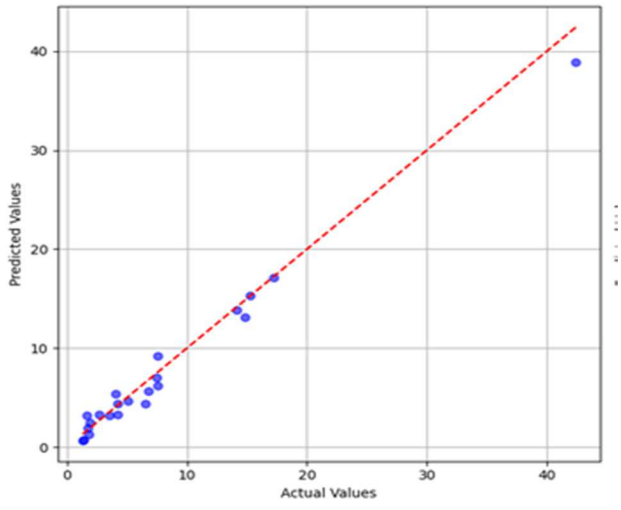


Fig.3 Actual vs. Predicted for Pre-monsoon

Fig. 3. Actual vs. Predicted values for the Pre-monsoon period. The scatter plot shows that most points lie close to the reference line, indicating good model performance with minor deviations at higher actual values.

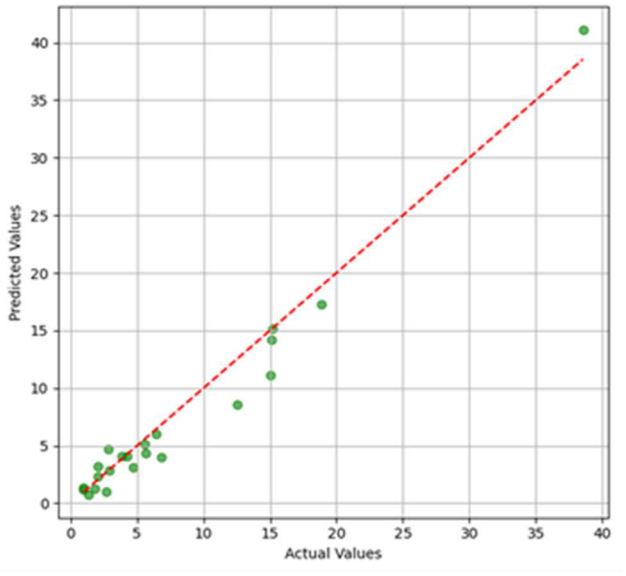


Fig.4 Actual vs. Predicted for Post-monsoon

Fig. 4. Actual vs. Predicted values for the Post-monsoon period. The scatter plot illustrates a similar trend, with predicted values aligning closely with actual observations, reflecting reliable accuracy across the season.

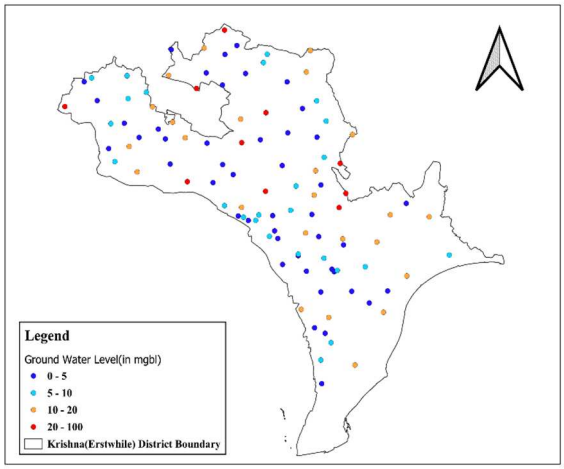


Fig. 5. Predicted Values: Pre-monsoon

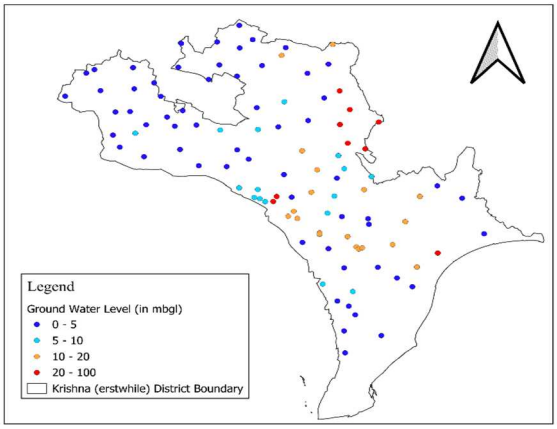


Fig. 6. Predicted Values: Post-monsoon

Fig. 5 and Fig. 6 show the spatial distribution of groundwater levels in Krishna (Erstwhile) district across different depth categories, highlighting areas with relatively low groundwater availability and potential depletion zones.

TABLE I. EVALUATION METRIC RESULTS

Metric	Value
RMSE (m)	2.34
MAE (m)	1.20
R^2	0.99

TABLE II. COMPARISON OF MODELS

Algorithm	RMSE	MAE	R^2
SVR	3.07	1.38	0.88
RF	2.90	1.35	0.89
GB	2.34	1.20	0.99

V. CONCLUSION AND FUTURE WORK

This study applied machine learning to predict groundwater levels and identify depletion zones in the erstwhile Krishna District of Andhra Pradesh. Gradient Boosting outperformed Random Forest and SVR and was used to forecast pre-monsoon and post-monsoon groundwater levels. Results highlighted seasonal variations and areas with deeper water tables, particularly before the

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monsoon. QGIS-based visualization demonstrated the value of data-driven insights for proactive water management.

Future work can include additional hydrogeological and climatic variables, extend predictions beyond 2024, and incorporate interactive visualization tools to support decision-making. The study provides a scalable framework for localized groundwater prediction and sustainable resource planning.

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